

MATERIAL SAFETY DATA SHEET

1.IDENTIFICATION OF THE SUBSTANCE / PREPARATION AND THE COMPANY

2,4-D DIMETHYLAMMONIUM SALT 840g/L SL

SUPPLIER NAME: ADAMA Huifeng (Jiangsu) Ltd.
SUPPLIER ADDRESS: Weier Road South Area of Ocean Economic Development Zone Dafeng Jiangsu Province 224145
SUPPLIER PHONE: +86-515-85055601
CHEMICAL NAME: 2,4-D Dimethylamine salt
CAS NUMBER: 2008-39-1
MOLECULAR FORMULA: $C_{10}H_{13}Cl_2NO_3$
MOLECULAR WEIGHT: 266.1
CHEMICAL FAMILY: Phenoxy compound
USE: Herbicide

2.COMPOSITION / INFORMATION ON INGREDIENTS

2,4-D Acid(present as the dimethylamine salt) 840g/L.
Water and other ingredients to 100%

3. HAZARDS IDENTIFICATION

The International Agency for Research on Cancer (IARC) lists chlorophenoxy herbicides in its Group 2B (limited evidence for carcinogenicity in humans.) The US EPA has given the chlorophenoxy herbicides 2,4-D, 2,4-DP, MCPP, and MCPA a Class D classification (not classifiable as to human carcinogenicity.) More current 2,4-D lifetime feeding studies in rats and mice did not show carcinogenic effects and a recent World Health Organization (WHO) review of 2,4-D toxicology has concluded that 2,4-D is not a carcinogen.

4. FIRST - AID MEASURES

INHALATION: If ill-effects occur, move to fresh air.
SKIN CONTACT: Flush area with large quantities of water.
EYE CONTACT: Flush with water for 15 minutes.
INGECTION: If swallowed, Call a doctor or get medical attention. Do not induce vomiting. Drink promptly a large quantity of milk, egg whites, gelatin solution, or if these are not available, drink large quantities of water. Avoid alcohol.
NOTES TO DOCTOR: No specific antidote.

5. FIRE FIGHTING MEASURES

FLASHPOINT: NA
FLAMMABLE LIMITS: NA
EXTINGUISHING MEDIA: Dry chemical, water spray (fog), foam or carbon dioxide
THERMAL DECOMPOSITION PRODUCTS: Nil
SPECIAL FIRE FIGHTING METHODS: Vapours may be toxic. Self contained breathing apparatus required. Do not allow contaminated runoff to enter drains.
UNUSUAL FIRE OR EXPLOSION HAZARDS: None

6. ACCIDENTAL RELEASE MEASURES

Use appropriate protective equipment. Large spill: Sweep the spill to a waste container.

7. HANDLING AND STORAGE

Precautions to be Taken: Do not contaminate water, food, or feed by storage or disposal. STORAGE: Store in original container in a locked storage

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area inaccessible to children or pets. Keep from freezing.

Other Precautions: Containers over 1 gallon and less than 5 gallons: Persons engaged in open pouring of this product must also wear coveralls or a chemical resistant apron. Containers of 5 gallons or more: Do not open-pour from this container. A mechanical system (such as probe and pump or spigot) must be used for transferring the contents of this container. If the contents of a non-refillable pesticide container are emptied, the probe must be rinsed before removal.

8. EXPOSURE CONTROLS / PERSONAL PROTECTION

Ventilation Requirements: Good local ventilation is required. Mechanical exhaust may be necessary if product is applied in an enclosed area.

Personal Protective Equipment: Clothing Requirement Statement: When mixing, loading or applying this product or repairing or cleaning equipment used with this product, wear eye protection (face shield or safety glasses), chemical resistant gloves, long-sleeved shirt, long pants, socks and shoes. It is recommended that safety glasses include front, brow and temple protection. For aerial applicators in an enclosed cockpit and applicators applying this product water is present or to intertidal areas below the mean high water mark. from a tractor that has a completely enclosed cab, eye protection is not required.

Personal Hygiene Statements: Wash hands, face and arms with soap and water as soon as possible after mixing, loading or applying this product. Wash hands, face and arms with soap and water before eating, smoking or drinking. Wash hands and arms before using toilet. After work, remove all clothing and shower using soap and water. Do not reuse clothing worn during the previous day's mixing and loading or application of this product without cleaning first. Clothing must be kept and washed separately from other household laundry. Remove saturated clothing as soon as possible and shower.

Respiratory Protection: If exposure limits may be exceeded, wear a dust/mist filtering respirator.

9. PHYSICAL AND CHEMICAL PROPERTIES

APPEARANCE:	Clear yellow liquid
ODOUR:	Slight ammoniacal odour
BOILING POINT:	>100°C (for water)
MELTING POINT:	less than 0°C
PH:	7-9
VAPOR PRESSURE:	16mmHg for water, 2,4-Ddimethylamine Salt is non volatile
FLASH POINT:	Not flammable
OTHER INFORMATION:	Completely soluble in water

10. STABILITY AND REACTIVITY

CHEMICAL STABILITY: Relatively stable in neutral, weakly acidic and weakly alkaline media; hydrolyzed by stronger acids and basis.

STORAGE STABILITY: Stable under normal storage conditions in original container for at least two years.

COMPATIBILITY: Compatible with most herbicide formulations except very strong acid/alkaline formulations

HAZARDOUS DECOMPOSITION PRODUCTS: None

11. TOXICOLOGICAL INFORMATION(2,4-D Acid)

ACUTE ORAL LD ₅₀	681mg/kg in the rats
ACUTE DERMAL LD ₅₀	>2150 mg/kg in rats
INHALATION LC ₅₀	>1.79 mg/l in rats (4 hr exposure)
SKIN IRRITATION	Non-irritant (4 hr exposure)
EYE IRRITATION	Corrosive. Causes irreversible eye damage.
SENSITISATION	Skin sensitizer
CARCINOGENICITY	No evidence of carcinogenicity
MUTAGENESIS	2,4-D is not mutagenic or has low mutagenic potential
TOXICOLOGICAL CLASSIFICATION	Low toxicity

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12. ECOLOGICAL INFORMATION

This product is toxic to aquatic invertebrates. Drift or runoff may adversely affect aquatic invertebrates and nontarget plants. For terrestrial uses, do not apply directly to water, or to areas where surface water is present or to intertidal areas below the mean high water mark. Do not contaminate water when disposing of equipment washwaters.

13. DISPOSAL CONSIDERATIONS

PESTICIDE DISPOSAL: Pesticide wastes are toxic. Improper disposal of excess pesticide, spray mixture, or rinsate is a violation of Federal law and may contaminate groundwater. If these wastes cannot be disposed of by use according to label instructions, contact your state Pesticide or Environmental Control Agency or the Hazardous Waste Representative at the nearest EPA Regional Office for guidance.

CONTAINER DISPOSAL: Triple rinse (or equivalent). Then offer for recycling or reconditioning, or puncture and dispose of in a sanitary landfill, or incineration, or if allowed by state and local authorities by burning. If burned stay out of smoke.

14. TRANSPORT INFORMATION

Hazardous Substances, Liquid, N.O.S., 9, UN3082, PGIII, RQ (2,4-D)

15. OTHER INFORMATION

The data given here is based on current knowledge and experience. The purpose of this safety data sheet is to describe the products in terms of their safety requirements. The data does not signify any warranty with regard to the products' properties.

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